

Evolutionary Bias Identification with Embeddings

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Abstract. This paper introduces EBIE (Evolutionary Bias Identification with Embeddings), a new method to help tackle algorithmic bias in natural language processing (NLP) tasks. The method leverages the powerful representation of word embeddings through an evolutionary algorithm, focusing on classification tasks. EBIE monitors shifts in individual embedding dimensions over generations and, by tracking these dimensional changes, identifies which parts of the embedding are most responsive to changes performed by genetic operations. These insights reveal critical features that influence model decisions and expose latent biases embedded within NLP classifiers. Through correlation analysis between individual tokens and classification scores, EBIE uncovers systematic biases in model behavior, such as reliance on stereotypical markers and neglect of nuanced expressions. By uncovering these tendencies, our methodology provides actionable insights to refine model training, enhance fairness, and improve robustness. Its flexibility ensures broad applicability across various NLP tasks, offering a powerful and versatile framework for developing more equitable and transparent machine learning systems.

Keywords: Evolutionary Algorithms · Transformer Models · Natural Language Processing · Bias Identification

1 Introduction

The widespread use of artificial intelligence (AI) algorithms has induced governments and society to create specific legislation to deal with the decisions made by these algorithms. While AI models are required to explain their decisions, they continue to grow in complexity, with some deep neural networks being scaled to trillions of parameters [5].

Together with explainability comes the problem of algorithmic bias, i.e., “unfair” outcomes produced by the model due to systematic errors. These errors usually come from shortcomings in the training data, which may have inherent

human bias, insufficient size, or skewed distribution. Understanding the models’ shortcomings is an essential and challenging task.

This paper introduces a new method to help tackle this problem in natural language processing (NLP) tasks. The method, EBIE (Evolutionary Bias Identification with Embeddings), leverages the powerful representation of word embedding with an evolutionary algorithm. We focus on classification tasks, although the framework is sufficiently generic to be applied to various NLP tasks. Starting from a list of random words, where each word is represented by an embedding, we evolve embedding vectors to maximize the results of a classification task.

By evolving the embedding vectors to obtain the best possible classification score, we reveal which words and sentences are most associated with the model output. This set of words, as discussed later, can help us understand the biases of the classifier without opening the black box or having access to the training dataset. At the same time, it can give hints on training data that might have been missing during the training process or insights into errors made by specific terms.

The rationale behind the proposed method is to monitor shifts in individual embedding dimensions over generations. By tracking these dimensional changes, we can identify which parts of the embedding are most responsive to changes performed by genetic operations. We measure changes across generations by mapping distances between top-performing individuals and their offspring, providing insights into how specific dimensions consistently shift in response to data classification. This approach helps reveal any inherent biases, offering a view into how particular embedded dimensions correlate with the target classes.

By using a metric of classification quality to drive the evolution process we can, for example, identify patterns in language that the model favors – represented by sentences that achieve a high classification score. Conversely, we can evolve sentences to receive lower scores, revealing text features the model penalizes. This is paramount, as historical examples have shown that machine learning models can unintentionally perpetuate discrimination based on factors such as race or gender.

Our contributions are twofold: we develop an evolutionary-based approach to identify bias in classification models by looking at the relationship between word embeddings and their impact on the output of classification systems. As a consequence, we recognize the embedding dimensions correlated to variations in fitness and understand the impact they may have on algorithmic bias.

2 Related Work

The growing adoption of machine learning in critical decision-making systems has raised significant concerns about fairness and bias, especially in black-box models. As these models often lack transparency, there is a pressing need for methodologies that can detect, interpret, and mitigate biases. Mehrabi [10] discusses how biases can originate from various sources, including data representa-

tion, measurement inaccuracies, and algorithmic design. Moreover, the feedback loop between data, algorithms, and users can perpetuate and amplify existing biases.

Similarly, the survey by Caton and Haas [1] underscores the complexity of ensuring fairness in machine learning, highlighting a structured taxonomy of pre-processing, in-processing, and post-processing interventions to address bias. It also emphasizes the challenges of balancing fairness with other performance metrics, such as accuracy, and explores the evolution of fairness concepts across different ML contexts. Evans et al. [3] emphasized that interpretable machine learning techniques can contribute to identifying and observing bias in models. The article further proposed a genetic programming-based method to extract simplified representations from complex black-box models. This technique applies multi-objective GP to balance accuracy with model transparency, similar to [7], which has the same objectives without the GA.

The study [12] is related to identifying bias in the context of NLP, providing a structured approach to detecting and analyzing biases in textual data. This work highlights the potential for biases in NLP models to perpetuate stereotypes or discrimination and offers a systematic methodology to identify such biases. The framework evaluates how specific word choices and structures impact model predictions, providing insights into mitigating biased outcomes. This aligns with the growing need for fairness and accountability in AI-driven systems.

Recent studies have begun to draw connections between large language models (LLMs) and evolutionary algorithms (EAs), illustrating shared principles of sequential processing, fitness shaping, and reproduction. Wang et al. [2] explore these parallels, providing a framework that highlights the interdisciplinary potential of LLM-enhanced EAs.

Fan et al. [4] proposed an approach to fairness testing in AI models using genetic algorithms. This method identifies scenarios where changes in sensitive attributes, such as gender or race, result in significant shifts in model outputs. By combining counterfactual explanations with genetic optimization techniques, this approach systematically explores biases embedded in models. This methodology highlights the effectiveness of genetic algorithms as tools for fairness audits, aligning with EBIE’s objective of uncovering latent biases in classification models.

CERTIFAI [13] and LORE (LOcal Rule-based Explanations) [6] are both frameworks that generate counterfactuals of a synthetic neighborhood utilizing genetic algorithms. Their application for black-box models as well as the quality of their model explanations are desirable properties for a variety of applications. In the case of CERTIFAI, it also quantifies the differences in counterfactual distances between groups to determine whether the model imposes disproportionate effort on certain groups to achieve favorable outcomes, measuring fairness and identifying potential biases through the “burden” metric. But, unlike EBIE, they require existing data to function, and their applicability in NLP may be constrained. Some commonly used local-explanation frameworks such as SHAP [9], also see limited usage when dealing with LLMs due to the unintuitiveness and

complexity of the explanations, as well as the usage of computational power for this domain [15]

Despite the many proposed methods for identifying biases, not much work has been done through the usage of evolutionary algorithms as previously presented. As these algorithms have been shown to be a promising tool to achieve explainability [11], we decided to evaluate its effectiveness specifically in the context of fairness and bias.

2.1 Research Gap

Despite advancements in addressing bias in AI, significant gaps remain in understanding how biases emerge and persist within classifiers, particularly at a lower level. Current approaches, such as attention visualization and saliency maps [14], provide surface-level insights but fail to probe the latent representations that drive a model’s decisions. These limitations leave many biases undetected or poorly understood.

While evolutionary algorithms have demonstrated promise in other domains for improving model fairness and adaptability, their application in NLP, specifically for bias detection, is still underexplored. Few tools exist that can dynamically manipulate a model’s embeddings to expose how linguistic patterns and features contribute to biased outcomes.

This gap highlights an opportunity to leverage genetic algorithms for evolving sentences, allowing researchers to uncover and analyze hidden biases within NLP models. By exploring these latent dimensions, with EBIE, we can develop a deeper understanding of models and training data, enabling targeted interventions to mitigate issues relating to its biases.

3 EBIE

The main goal of the proposed method, EBIE, is to evolve a sequence of words that maximizes the quality of the classification model being “audited”. The model is audited in the sense that the method provides insights into biases that can emerge within it, identifying patterns in word choices and phrase structures that are more commonly associated with particular classifier outcomes. Figure 1 illustrates the evolutionary process. Having a population of phrases, where each phrase is represented by a list of embeddings randomly generated E_R , we evaluate the “strength” of the contribution of E_R to the output of the model $\mathcal{M}$. For example, assume our classifier outputs whether a phrase/sentence is positive or negative, i.e., $c \in \{+, -\}$. The model will return a probability $P(c|E_R)$ that will be used to guide the evolutionary process, where phrases that induce higher model probabilities are more favored.

This probability guides the evolutionary process in the selection phase, followed by a crossover, mutation and two new operators that add words to or remove words from current phrases. This process is repeated until a predefined set of generations is met. This process creates sentences that somehow mirror what existed in the training set.

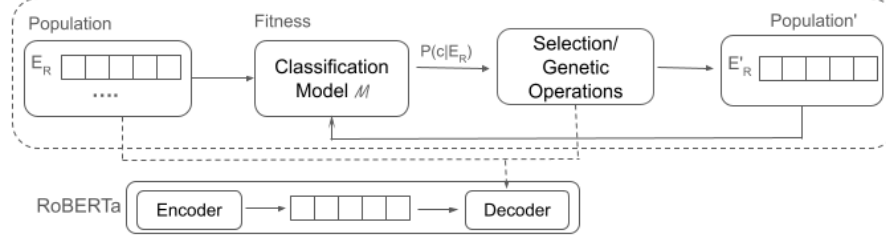


Fig. 1: Embeddings Representation Evolutionary Process.

3.1 Representation and Population Initialization

Each individual represents a sentence, and each word of the sentence is coded by using an embedding vector produced by a pre-trained model. However, recall that transformer models do not encode words directly, but tokens. A token can be a complete word or a subset of a word. For the sake of simplicity, we use the terms “words” and “tokens” interchangeably. We used the RoBERTa model [8] to convert embeddings into words. This model was chosen due to its high accuracy, extensive pre-training, and widespread use in the NLP community. By following a pre-trained model, the manipulations performed on the embedding vectors by the genetic operators will retain, to an extent, semantic meaning and coherence, ensuring that the evolved phrases remain meaningful.

Each individual E is represented by a list of word embeddings e_i , where i is the number of words of the sentence and can vary from 1 to $size_p$. Each word embedding vector has a size n , and hence the length of the individual vector representation can vary from n , when the individual represents a sentence with a single word, to $n \times size_p$, when an individual represents a sentence with the maximum number of words allowed. In this work, n has size 1024.

Having the representation, we need to initialize the population of embeddings. For the first population, we randomly selected tokens from the RoBERTa model and initialized each individual as a single token. By initializing tokens randomly in the embedding space, we wanted to promote diversity in the initial population. This diversity is crucial, as the evolutionary operators involve gradual shifts within the embedding space. A diverse set of points enhances the exploration potential of the algorithm, increasing the likelihood of discovering novel and diverse phrase structures during the evolutionary process.

3.2 Fitness

As mentioned, the fitness function will use the classification model we want to explore. Here we work with a multi-class emotion classifier. This choice is motivated by the intuitive nature of emotion classification, which facilitates qualitative analysis and proof of concept. The simplicity of the task allows for a clearer illustration of our approach compared to more complex tasks.

Importantly, the specific model employed is irrelevant to the methodology itself. Our approach is designed to work with any classification model that works with text. The chosen model serves only as a practical mean to demonstrate the flexibility and applicability of the methodology, ensuring that its core principles remain universally adaptable across various tasks and domains.

We selected the “j-hartmann/emotion-english-distilroberta-base”¹, a multi-class emotion classifier, as our model $\mathcal{M}$. This model is a fine-tuned version of DistilRoBERTa, which is a distilled version of the RoBERTa base model. The fine-tuning process trains DistilRoBERTa to classify sentences into one of seven emotion categories: Joy, Sadness, Anger, Disgust, Fear and Surprise. The model was fine-tuned with 19,677 examples and has a vocabulary of 50,265 tokens. It outputs the probability for each one of the seven possible classes.

Therefore, for a given individual E that represents a list of embeddings that maps to a phrase, we look at the probability returned by $\mathcal{M}$ when analyzing a specific class c . This makes understanding the impact of words/phrases in a classifier easier. Hence, each individual is given a probability score that represents the probability $P(c|E_R)$ of the phrase it represents falling within the target class c being analyzed. This score serves as our fitness measure, with higher-scoring phrases having a greater probability of selection for crossover and mutation in subsequent generations. Through this process, the evolution is directed towards linguistic configurations that maximize the expression of the desired target category, revealing the biases and helping to understand the inner workings of black-box models.

3.3 Evolutionary operators

One of the key challenges of this work was determining how to crossover and mutate these embeddings while maintaining the ability to convert them back into coherent words.

For the crossover operation, given two selected individuals E_1 , E_2 , if the number of tokens in E_x or $E_y > 1$, one of the e_r tokens in E is randomly selected. Then, for each dimension in the selected embeddings, values can be swapped to produce new offspring according to a given probability. This method aims to diversify the embeddings, introducing variation in their abstract representations while preserving a degree of consistency with the original word meanings.

For mutation, given an individual E_x , we introduced controlled random noise into one randomly selected embedding e_r . Specifically, we applied a perturbation mechanism where each dimension of the embedding vector was either increased or decreased by 10%. This process changed the vector’s position in the embedding space, effectively modifying the semantic meaning of the word in a controlled yet random direction.

After mutation and/or crossover, the resulting embeddings may no longer correspond directly to a specific word or token in the model’s vocabulary. Therefore, the next step involves converting these altered embeddings back into coherent

¹ <https://huggingface.co/j-hartmann/emotion-english-distilroberta-base>

words. For that, we used the intrinsic mechanisms employed during the pre-training phase of RoBERTa.

RoBERTa uses a training mechanism that masks a subset of input tokens and learns to predict these masked tokens by generating embeddings for each of them. These embeddings are processed by attention mechanisms, which analyze contextual relationships within the input sequence. Using this context, the model predicts the most likely tokens, outputting a probability distribution over its vocabulary.

By applying this mechanism to our modified embeddings, we can map them back to plausible words or tokens. Specifically, we used the model’s language generation head to generate a ranked list of probable words. From this list, we sampled the top-50 candidates to ensure the resulting text remained semantically meaningful and diverse. This process preserves the coherence of the evolved phrases while allowing for variations introduced by the genetic algorithm.

This method allows us to exploit the continuous nature of the embedding space while still extracting semantically coherent words. It also ensures the evolutionary algorithm produces valid, interpretable text while maintaining the diversity introduced through evolutionary operations.

To further enhance the evolution of the phrase, we introduced two new operators that probabilistically add new/remove tokens to/from individuals, enabling the generation of complete and more complex sentences for deeper analysis. New tokens were introduced into an individual similar to the way the population initialization was performed. Specifically, we filled the dimensions of a new token embedding at random, and then sampled from the top most probable words using the same probabilistic approach based on the predicted token distribution. As generations progressed, new words were incrementally added to sentences, with the maximum number of tokens in a sentence being capped by the number of generations. These two operators maintain a balance between growth and brevity in sentence structure and let the model being evaluated decide the “right” number of tokens in a sentence.

In essence, this methodology allows us to evolve sentences that “fit” a specific model’s preferences or biases, offering a direct and dynamic way to explore the inner workings of any NLP classification model. By iteratively evolving text and using the target model’s feedback as the evolutionary guide, we are not only able to optimize sentence structures for high or low classification scores but also provide a structured approach to uncover the hidden biases and decision patterns within the model.

4 Experimental Analysis

The proposed methodology was tested using two emotions that can be predicted by the classification model: Joy and Sadness. For each of the analyzed emotions, we conducted 30 independent runs of the EA. Each run considered a population size of 100 evolved for 200 generations. A tournament selection with $k = 2$ was used, and the probabilities of applying crossover and mutation operators

were defined as 0.8 and 0.2, respectively. For crossover, there was a 0.02 chance of swapping each dimension within the embedding. The probabilities of both adding or deleting tokens from the individuals were set to 0.3. These values were determined empirically, as the novelty of the methodology provides limited opportunities for direct comparison with existing approaches. Refinements and studies of the impact of these parameters are left for future work.

This section is divided into two parts. First, we show the metrics and insights we can derive from the model based on the sentences generated by the evolutionary algorithm. Next, we analyze how we can qualitatively modify a sentence to change its bias based on our understanding of the model.

4.1 Tokens Evolution Results

We start by analyzing how tokens evolve during the evolutionary process. Figure 2 presents the mean difference between the embeddings of the individuals of the first generation and those of the last generation across each of the 1,024 embedding dimensions when analyzing the “Joy” sentiment category. This approach highlights specific dimensions within the embedding space that exhibit greater variability from random vectors and the most optimal ones, suggesting that they play a more prominent role in driving the model’s output changes.

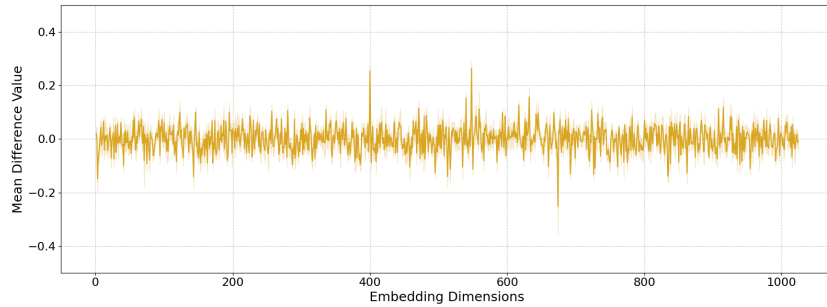


Fig. 2: Line graph showing the mean difference in “Joy” sentence embeddings between the first and last generations, measured across each embedding dimension.

In transformer-based language models, each embedding dimension theoretically encodes a specific characteristic or semantic feature of the word or token it represents. By identifying the dimensions most responsive to evolutionary changes, such as dimensions 548, 400, and 674 in Figure 2, we can infer which aspects of the embedding space are most influential in determining the classification outcome. Changes in these dimensions might correspond to linguistic features like positivity, tone, or context, which the model heavily weights when predicting the “Joy” sentiment.

This insight is critical for several reasons. First, it offers a perspective on the latent structure of the model’s learned representations, allowing researchers to

identify which semantic attributes are prioritized by the classifier. Second, understanding these influential dimensions can guide targeted interventions, such as fine-tuning or modifying training data to address biases or improve classification robustness. For example, if certain dimensions disproportionately impact specific sentiments, these dimensions could be regularized or adjusted to ensure more balanced decision-making.

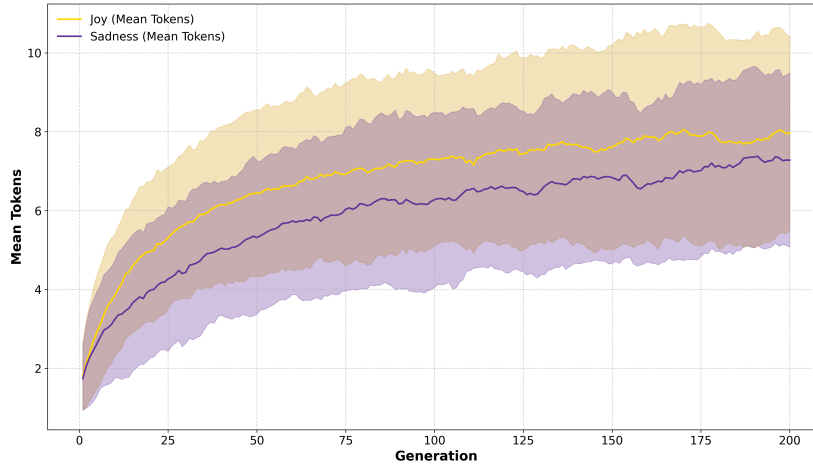


Fig. 3: Mean number of Tokens per generation in “Joy” and “Sadness” Categories. The yellow line represents the mean token length for “Joy”, while the purple line represents “Sadness”. The shaded areas indicate the range of variation across generations, reflecting the variability in sentence lengths for each emotion.

Furthermore, we analyzed the median number of tokens the model predicts as ideal. Figure 3 shows, for each generation, the average number of tokens used by the phrases in the “Joy” (yellow) and “Sadness” (purple) categories across multiple runs. Note that both categories exhibit a rapid increase in token count during the initial generations (0-50), suggesting the evolutionary algorithm initially adds tokens to explore more complex sentence structures that might better capture each sentiment. After this phase, both categories reach a plateau, with average sentence lengths stabilizing between 6 and 10 tokens as an ideal phrase length for this model to capture the targeted emotions.

It is notable that, despite the similarities in curves, the “Joy” category consistently exhibited a higher average token count compared to “Sadness” across generations. This observation suggests that joyful sentences tend to be more complex, potentially reflecting characteristics of the training data. These findings underscore how the evolutionary algorithm adapts sentence structures based on sentiment, revealing underlying biases and preferences in the model’s handling of emotional expressions.

Figure 4 shows, for one individual, how the embeddings have evolved from the first to the last generation. In this run, we set the probabilities for adding or removing tokens to 0, and individuals will always have a single token. For every 20 generations, we show the decoded words so we can qualitatively assess the tokens actually evolve to improve the scores for class “Joy”.

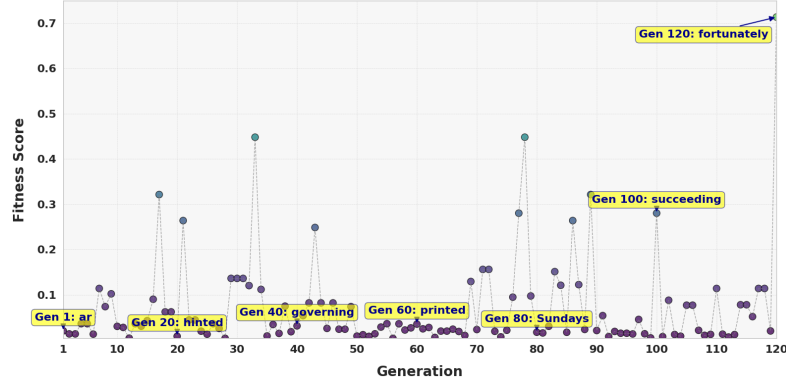


Fig. 4: Example of individual evolved over the generations.

A qualitative analysis of the evolved words was also performed. As shown in Figure 5, we can observe which trends introduce a particular bias toward each classification. In the left word cloud, we see trends associated with “Joy”, where expected words such as “fun”, “enjoy”, and “cheerful” are prominent. Similarly, in the “sadness” category, words like “sadly”, “grief”, and “unhappy” appear most frequently.

Interestingly, while these trends align with expected associations, they seem almost stereotypical of these categories. More complex words or phrases, as well as negations, are noticeably absent in these word clouds. This suggests that the model might not favor more nuanced or intricate sentence constructions, a topic we will explore further in the next section. This raises questions about the model’s depth of understanding and its potential limitations in recognizing varied expressions within these emotional categories, even though they seem to align with our expectations on a surface level.

We have also analyzed the correlation between individual tokens evolved by the algorithm and their likelihood of leading to a classification of “Joy” or “Sadness”, as illustrated in Figure 6. In this analysis, each token was evaluated independently rather than as part of a full sentence, allowing us to determine which tokens maximize or minimize the model’s fitness function for each category. The Pearson correlation coefficients for tokens associated with the classification of “Sadness”, shown on the left side of Figure 6, reveal terms that either increase or decrease the likelihood of a sadness label. Words with strong positive correlations, such as “canceled”, “poignant”, “death”, and “sad” significantly increase the



Fig. 5: Word clouds illustrating model-driven linguistic associations for “Joy” and “Sadness” classifications. The left figure shows terms associated with “Joy”, while the right figure displays terms related to “Sadness” sentiment.

probability of the model’s output for sadness. These terms commonly represent themes of loss, suffering, or other negative outcomes, aligning with sadness-related contexts within the training data. Interestingly, “hearing”, typically a neutral term, also shows a high positive correlation with sadness, which may be due to the word’s frequent appearance in legal contexts (e.g., “court hearing”), which often carry negative connotations. Conversely, words with negative correlations to sadness, like “question”, “page”, “email”, and “interest” are neutral or lack inherent emotional content, reducing the probability of the model classification as sadness. This distribution suggests the model may rely heavily on explicit sadness markers, potentially limiting its ability to detect more nuanced expressions of negative sentiment.

For the sentiment of “Joy”, shown on the right side of Figure 6, highly correlated words, such as “fun”, “laughably”, “hilarious”, “love”, and “great!” are positive and align with stereotypical markers of joy. These words convey unambiguous positive sentiments, which the model interprets as joyful. In contrast, words negatively correlated with joy are more neutral, formal, or question-oriented, which lowers the probability of a joy classification. Terms like “why” and “what” may imply ambiguity or skepticism, which the model may interpret as neutral or slightly negative. Additionally, words like “advertisement” and “public” often appear in more formal contexts, making them less likely to be associated with joy. This pattern suggests the model struggles to identify subtle or context-dependent expressions of joy, as it leans heavily on direct positive indicators.

In sum, both analyses indicate that the model relies on stereotypical markers – favoring overtly positive or negative terms, which limits its ability to capture complex or nuanced emotional cues. This bias toward direct emotional indicators highlights a potential area for improvement in the model’s training data and interpretive algorithms.

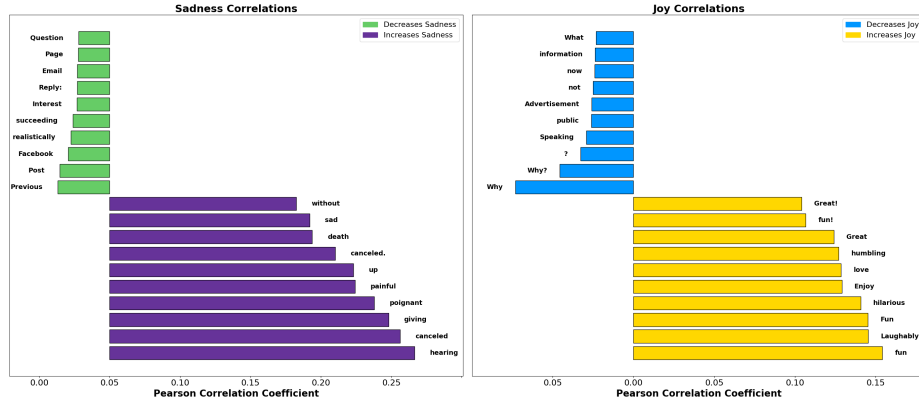


Fig. 6: Bar charts showing the Pearson Correlation between sentence tokens. In the left figure, green bars indicate terms that decrease “Sadness” category, and purple bars indicate terms that increase “Sadness”. In the right figure, blue bars indicate terms that decrease the “Joy” category, while the yellow ones indicate terms that increase “Joy”.

#Tokens	Sentences	$p(c=\text{joy} \text{sentence})$
5	“I am happy”	0.9657
8	“I am extremely happy right now”	0.9799
8	“I am happy I am happy”	0.9672
11	“I am extremely happy right now today is amazing”	0.8757
11	“I am happy I am happy I am happy”	0.9602

Table 1: Sentences with varying token lengths and their probabilities of being classified as “Joy”.

4.2 Practical Uses

The findings from the previous section can now be applied to real sentences, providing insights into the model’s biases in practical contexts. Table 1 presents some examples. Our goal was to understand how the average token range of 6–10, identified earlier, translates into real sentences. Hence, we started with a base sentence of 5 tokens and expanded it in two ways: by repeatedly copying the same sentence and by adding more varied, joyous content.

This process produced two sentences with 8 tokens and two with 11 tokens. The results clearly show that sentences with 8 tokens achieve the highest scores, while those with 5 or 11 tokens fall short. This demonstrates the model’s sensitivity to token length and its tendency to favor sentences within a specific range.

The next straightforward approach is to observe how the most prominent words observed in the word cloud behave within the text. In previous paragraphs, we discussed how the model’s word choices seemed somewhat stereotypical, pos-

Sentence	Score
“This is amazing, i feel like I have never felt any sadness”	0.1145
“I was on cloud nine this morning”	0.0144
“Tonight I was over the moon”	0.0252

Table 2: Real sentences and their corresponding “Joy” scores, showcasing the model’s difficulty with negations and idiomatic expressions.

sibly due to the challenge of processing more complex contexts. We hypothesized that the model may struggle with negations or more intricate sentences, which has proved to be accurate. For example, in cases of negation, such as the first sentence in Table 2, the model fails to associate negations of sadness coupled with a joyous context with “Joy”, despite the evident relationship. Moving to the second and third sentences, it is evident that the model struggles with idiomatic expressions. When joy is conveyed indirectly, without explicitly joyful words, the model fails to grasp the intended meaning.

Figure 6 illustrates how certain factors negatively impact the categories, leading to a lower score. For “Joy”, note that phrasing a statement as a question can significantly reduce its score. This trend is particularly noticeable in Table 3. For instance, when a generally positive, happy statement is rephrased as a rhetorical question – maintaining the same intended meaning – the model consistently interprets the question as significantly less joyful.

Sentence	Joy Score
“Life is so perfect right now”	0.9508
“How can life be so perfect right now?”	0.0289

Table 3: Scores of affirmative versus interrogative sentences of “Joy”.

We performed the same analysis for the “Sadness” category. As shown in Figure 6, the model tends to downplay sentences related to social media or the Internet. This result is unexpected but proves accurate when tested. Table 4 presents the same sentence twice, with the second version preceded by an internet-related context. This context dramatically lowers the score, highlighting again some of the biases present in the model.

Sentence	Sadness Score
“I have no one in my life now”	0.9242
“She never replied my email, I have no one in my life now”	0.3877

Table 4: Differences in scores of a sad sentence and the same sentence but with Internet-related content.

Building upon the analysis, we leveraged our understanding of the dimensions that, on average, contribute most significantly to fitness. We generated 100 random tokens and evaluated them in the “Joy” category. Then, we selectively modified the values of the embeddings by the mean difference (shown in Figure 2) to the top 20, 10, and 3 most influential dimensions, affecting at most 2% of the original dimensions. We measured the scores of these modified embeddings, and the results show improvements of 14.29%, 122.81%, and 187.80% when modifying the top 3, 10 and 20 dimensions, respectively. The data demonstrates that even with minimal changes the scores significantly improve. This improvement occurs regardless of the specific words the embeddings might decode, emphasizing the impact of these key dimensions on the model’s output.

5 Discussion and Conclusions

The results demonstrate a powerful method to uncover biases within NLP classification models. By leveraging evolutionary algorithms with an expansive search space – encompassing all possible sentences – and using the classification model as the fitness evaluator, the evolved sentences gravitate towards patterns and structures that reflect the model’s training data. This alignment provides a unique lens through which we can analyze and deconstruct model behavior, offering an “x-ray effect” on the training set and the model’s inherent preferences.

This becomes even more interesting when we consider our method does not rely on any external data, which differs it from almost any existing methodology. It operates without access to training datasets and uses only the model as the evaluator. This makes it versatile and applicable across a wide range of classifiers.

Even for models that are not embedding-based, the fitness function can still guide the evolution of sentences, enabling the same kind of analysis. This flexibility means the methodology is not tied to a specific model architectures or tasks, making it a powerful tool for uncovering biases and understanding the classifiers.

Given our results, an obvious next step is using EBIE and the analysis above to effectively reduce model bias. When a trained model’s behavior is challenging to interpret, this methodology can pinpoint areas where the model falls short and where it concentrates most of its “attention”.

For example, the earlier analysis shows that insights into common words, embeddings, and token counts could guide data diversification. By expanding and modifying the training phrases based on these findings, we could introduce more varied and balanced data, ensuring the model becomes less reliant on specific patterns and more robust in handling diverse inputs. This approach offers a practical pathway to improve both the fairness and performance of any classifier.

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